

Week	Date	Task Description	Time Spent	Special Problems Faced or Difficulties Encountered	Action Taken	Learning Outcomes
Week 1	Dec. 15, 2024	Analysis of Current Dataset Recommender Systems and Identification of Loopholes	6 h	- Difficulty finding datasets with metadata that is culturally and linguistically diverse.	- Used Kaggle and Hugging Face to search and collect a variety of datasets that represent different languages and cultures.	- Gained an understanding of the challenges in dataset diversity and metadata compatibility across languages.
		Study of Current Multilingual Embedding Models: Sentence-BERT, Hugging Face Transformers		- Difficulty understanding how embeddings work across multiple languages, especially for context alignment.	- Studied various research papers that discussed cross-lingual embeddings and their role in aligning multilingual context.	- Improved understanding of the role of embeddings in language understanding, especially in multilingual systems.
		Training Open Multilingual Evaluation and Testing Datasets		- Not enough datasets available for testing and evaluation, especially for underrepresented languages.	- Collected open datasets that include multiple languages to help test the recommendation engine.	- Learned the process of training multilingual models using open datasets and the importance of proper dataset evaluation for each language.
Week 2	Dec. 22, 2024	Engineered the Architecture of a Multilingual Recommendation Engine	8 h	- Variations in user queries across languages created issues with correctly interpreting intents.	- Created a flexible architecture that could handle multilingual data and process it in a way that the recommendation engine would understand.	- Gained insights into designing recommendation engines that are robust and adaptable to various languages and user inputs.
		Developed a Pre-processing Pipeline for User Queries in Multiple Languages		- Language-specific interpretations made it difficult to normalize queries effectively.	- Built a pre-processing pipeline that cleaned and normalized queries in a way that minimized language-specific issues, like synonym variations.	- Learned about effective ways to preprocess and normalize multilingual user input to make it consistent across languages.
		Experimented with Sentence Embeddings for Similarity Scoring		- Some language embeddings performed better than others, causing inconsistency in similarity scoring.	- Fine-tuned sentence embeddings and tested multiple variations to improve the quality of similarity scores across languages.	- Deepened understanding of how to work with embeddings for similarity scoring and the importance of fine-tuning models for different languages.
Week 3	Dec. 29, 2024	Integration of the Recommendation Engine into Vector DB RAG Operations	9 h	- Real-time query processing was slow, and there was a trade-off between speed and accuracy.	- Optimized the vector database queries and used more efficient search algorithms to improve query processing speed.	- Learned how to improve the performance of real-time query systems by optimizing database operations and enhancing similarity search speed.
		Evaluated System Performance with Test Queries in Hindi, English, Malayalam, and Kannada		- Low-resource languages like Malayalam did not provide sufficient data for effective recommendation performance.	- Increased dataset size by generating synthetic data for low-resource languages to ensure better recommendation coverage.	- Learned how to evaluate system performance and handle low-resource languages by supplementing data with synthetic data to improve accuracy.
		Metrics like Precision and Recall to Visualize Recommendation Accuracy		- Precision and recall measurements were difficult to balance, as certain languages had data sparsity, which affected the recall scores.	- Fine-tuned evaluation metrics based on specific language strengths and weaknesses, using precision-recall curves to assess performance.	- Gained hands-on experience in applying performance metrics (precision and recall) in real-world recommendation systems and understanding how to improve them.
Weekly Summary		Key Achievements	23 h			- Developed the core foundation for a multilingual recommendation system.
		- Designed and implemented the initial prototype of the dataset recommendation engine.				- Established a feedback loop for continuous improvement, critical in enhancing the recommendation engine's accuracy and effectiveness.
		- Set up a continuous pipeline to evaluate and improve recommendation accuracy.				- Gained practical experience in the iterative process of fine-tuning embeddings for specific languages to improve recommendation accuracy.
		- Built a framework for embedding fine-tuning based on language-specific data.				
		Challenges Encountered				- Realized the importance of refining query interpretation models and fine-tuning embeddings based on language.
		- Variations in how languages express meaning made it difficult to align embeddings and interpret queries consistently. - Balancing speed and accuracy was crucial for ensuring that the recommendation engine performs well in real-time applications.				- Understood how to fine-tune the system for real-time use cases, ensuring efficient query processing.